

▶ PSYCHIATRY—PSYCHOLOGY

Classical conditioning

Learning in which a natural response (salivation) is elicited by a conditioned, or learned, stimulus (bell) that previously was presented in conjunction with an unconditioned stimulus (food).

Usually elicits **involuntary** responses. Pavlov’s classical experiments with dogs—ringing the bell provoked salivation.

Operant conditioning

Learning in which a particular action is elicited because it produces a punishment or reward. Usually elicits **voluntary** responses.

Reinforcement	Target behavior (response) is followed by desired reward (positive reinforcement) or removal of aversive stimulus (negative reinforcement).	Skinner operant conditioning quadrants: <table><tr><td></td><th>Increase behavior</th><th>Decrease behavior</th></tr><tr><th>Add a stimulus</th><td>Positive reinforcement</td><td>Positive punishment</td></tr><tr><th>Remove a stimulus</th><td>Negative reinforcement</td><td>Negative punishment</td></tr></table>		Increase behavior	Decrease behavior	Add a stimulus	Positive reinforcement	Positive punishment	Remove a stimulus	Negative reinforcement	Negative punishment
	Increase behavior		Decrease behavior								
Add a stimulus	Positive reinforcement		Positive punishment								
Remove a stimulus	Negative reinforcement	Negative punishment									
Punishment	Repeated application of aversive stimulus (positive punishment) or removal of desired reward (negative punishment) to extinguish unwanted behavior.										
Extinction	Discontinuation of reinforcement (positive or negative) eventually eliminates behavior. Can occur in operant or classical conditioning.										

Transference and countertransference

Transference	Patient projects feelings about formative or other important persons onto physician (eg, psychiatrist is seen as parent).
Countertransference	Physician projects feelings about formative or other important persons onto patient (eg, patient reminds physician of younger sibling).

Ego defenses

Thoughts and behaviors (voluntary or involuntary) used to resolve conflict and prevent undesirable feelings (eg, anxiety, depression).

IMMATURE DEFENSES	DESCRIPTION	EXAMPLE
Acting out	Subconsciously coping with stressors or emotional conflict using actions rather than reflections or feelings.	A patient skips therapy appointments after deep discomfort from dealing with his past.
Denial	Avoiding the awareness of some painful reality.	A patient with cancer plans a full-time work schedule despite being warned of significant fatigue during chemotherapy.
Displacement	Redirection of emotions or impulses to a neutral person or object (vs projection).	After being reprimanded by her principal, a frustrated teacher returns home and criticizes her wife’s cooking instead of confronting the principal directly.
Dissociation	Temporary, drastic change in personality, memory, consciousness, or motor behavior to avoid emotional stress. Patient has incomplete or no memory of traumatic event.	A survivor of sexual abuse sees the abuser and suddenly becomes numb and detached.

Ego defenses (continued)

IMMATURE DEFENSES	DESCRIPTION	EXAMPLE
Fixation	Partially remaining at a more childish level of development (vs regression).	A college student continues to suck her thumb when studying for stressful exams.
Idealization	Expressing extremely positive thoughts of self and others while ignoring negative thoughts.	A patient boasts about his physician and his accomplishments while ignoring any flaws.
Identification	Largely unconscious assumption of the characteristics, qualities, or traits of another person or group.	A resident starts putting her stethoscope in her pocket like her favorite attending, instead of wearing it around her neck like before.
Intellectualization	Using facts and logic to emotionally distance oneself from a stressful situation.	A patient diagnosed with cancer discusses the pathophysiology of the disease.
Isolation (of affect)	Separating feelings from ideas and events.	Describing murder in graphic detail with no emotional response.
Passive aggression	Demonstrating hostile feelings in a nonconfrontational manner; showing indirect opposition.	A disgruntled employee is repeatedly late to work, but won't admit it is a way to get back at the manager.
Projection	Attributing an unacceptable internal impulse to an external source (vs displacement).	A man who wants to cheat on his wife accuses his wife of being unfaithful.
Rationalization	Asserting plausible explanations for events that actually occurred for other reasons, usually to avoid self-blame.	An employee who was recently fired claims that the job was not important anyway.
Reaction formation	Replacing a warded-off idea or feeling with an emphasis on its opposite (vs sublimation).	A stepfather treats a child he resents with excessive nurturing and overprotection.
Regression	Involuntarily turning back the maturational clock to behaviors previously demonstrated under stress (vs fixation).	A previously toilet-trained child begins bedwetting again following the birth of a sibling.
Repression	Involuntarily withholding an idea or feeling from conscious awareness (vs suppression).	A 20-year-old does not remember going to counseling during his parents' divorce 10 years earlier.
Splitting	Believing that people are either all good or all bad at different times due to intolerance of ambiguity. Common in borderline personality disorder. Borders split countries.	A patient says that all the nurses are cold and insensitive, but the physicians are warm and friendly.
MATURE DEFENSES		
Sublimation	Replacing an unacceptable wish with a course of action that is similar to the wish but socially acceptable (vs reaction formation).	A teenager's aggression toward her parents because of their high expectations is channeled into excelling in sports.
Altruism	Alleviating negative feelings via unsolicited generosity, which provides gratification (vs reaction formation).	A mafia boss makes a large donation to charity.
Suppression	Intentionally withholding an idea or feeling from conscious awareness (vs repression); temporary.	An athlete focuses on other tasks to prevent worrying about an important upcoming match.
Humor	Lightheartedly expressing uncomfortable feelings to shift the internal focus away from the distress.	A nervous medical student jokes about the boards.

Mature adults wear a **SASH**.

Grief

Natural feeling that occurs in response to the death of a loved one. Symptoms and trajectory vary for each individual, are specific to each loss, and do not follow a fixed series of stages. In addition to guilt, sadness, and yearning, patients may experience somatic symptoms, hallucinations of the deceased, and/or transient episodes of wishing they had died with or instead of their loved one. Typical acute grief is time limited (adaptations within 6 months) and is not a disorder.

Prolonged grief disorder—diagnosed if grief remains intense, persistent, and prolonged (at least 6–12 months), significantly impair functioning, is inconsistent with patient’s cultural or religious norms, and do not meet criteria for another disorder (eg, major depressive disorder [MDD]).

Normal infant and child development

Milestone dates are ranges that have been approximated and vary by source. Children not meeting milestones may need assessment for potential developmental delay.

AGE	MOTOR	SOCIAL	VERBAL/COGNITIVE
Infant	Parents	Start	Observing,
0–12 mo	P rimitive reflexes disappear— M oro, r ooting, p almar, B abinski (Mr. Peanut Butter) P osture—lifts head up prone (by 1 mo), rolls and sits (by 6 mo), crawls (by 8 mo), stands (by 10 mo), walks (by 12–18 mo) P icks—passes toys hand to hand (by 6–9 mo), P incer grasp (by 10 mo) P oints to objects (by 12 mo)	S ocial smile (by 2 mo) S tranger anxiety (by 6 mo) S eparation anxiety (by 9 mo)	O rients—first to voice (by 4 mo), then to name and gestures (by 9 mo) O bject permanence (by 9 mo) O ratory—says “mama” and “dada” (by 10 mo)
Toddler	Child	Rearing	Working,
12–36 mo	C ruises, takes first steps (by 12 mo) C limbs stairs (by 18 mo) C ubes stacked (number) = age (yr) × 3 C utlery—feeds self with fork and spoon (by 20 mo) K icks ball (by 24 mo)	R ecreation—parallel play (by 24–36 mo) R approchement—moves away from and returns to parent (by 24 mo) R ealization—core gender identity formed (by 36 mo)	W ords—uses 50–200 words (by 2 yr), uses 300+ words (by 3 yr)
Preschool	Don’t	Forget, they’re still	Learning!
3–5 yr	D rive—tricycle (3 wheels at 3 yr) D rawings—copies line or circle, stick figure (by 4 yr) D exterity—hops on one f oot by 4 yr (“ 4 on one f oot”), uses buttons or zippers, grooms self (by 5 yr)	F reedom—comfortably spends part of day away from parent (by 3 yr) F riends—cooperative play, has imaginary friends (by 4 yr)	L anguage—understands 1000 (3 zeros) words (by 3 yr), uses complete sentences and prepositions (by 4 yr) L egends—can tell detailed stories (by 4 yr)